Network Intrusion Detection System

Configure rules and alerts

Introduction

A Network Intrusion Detection System (NIDS) monitors network traffic to identify suspicious activities. It examines network packets and generates alerts when the traffic matches predefined detection rules. This project demonstrates the basic working of a NIDS using Snort in a Kali Linux virtual machine.

1. Objectives Understand the basic working of a Network Intrusion

Detection System.

Install and configure Snort.

Create a basic network traffic detection rule.

Monitor authorized lab traffic.

Observe and document detection alerts.

2. Tools and Requirements Kali Linux

VMware Workstation

Snort Intrusion Detection System

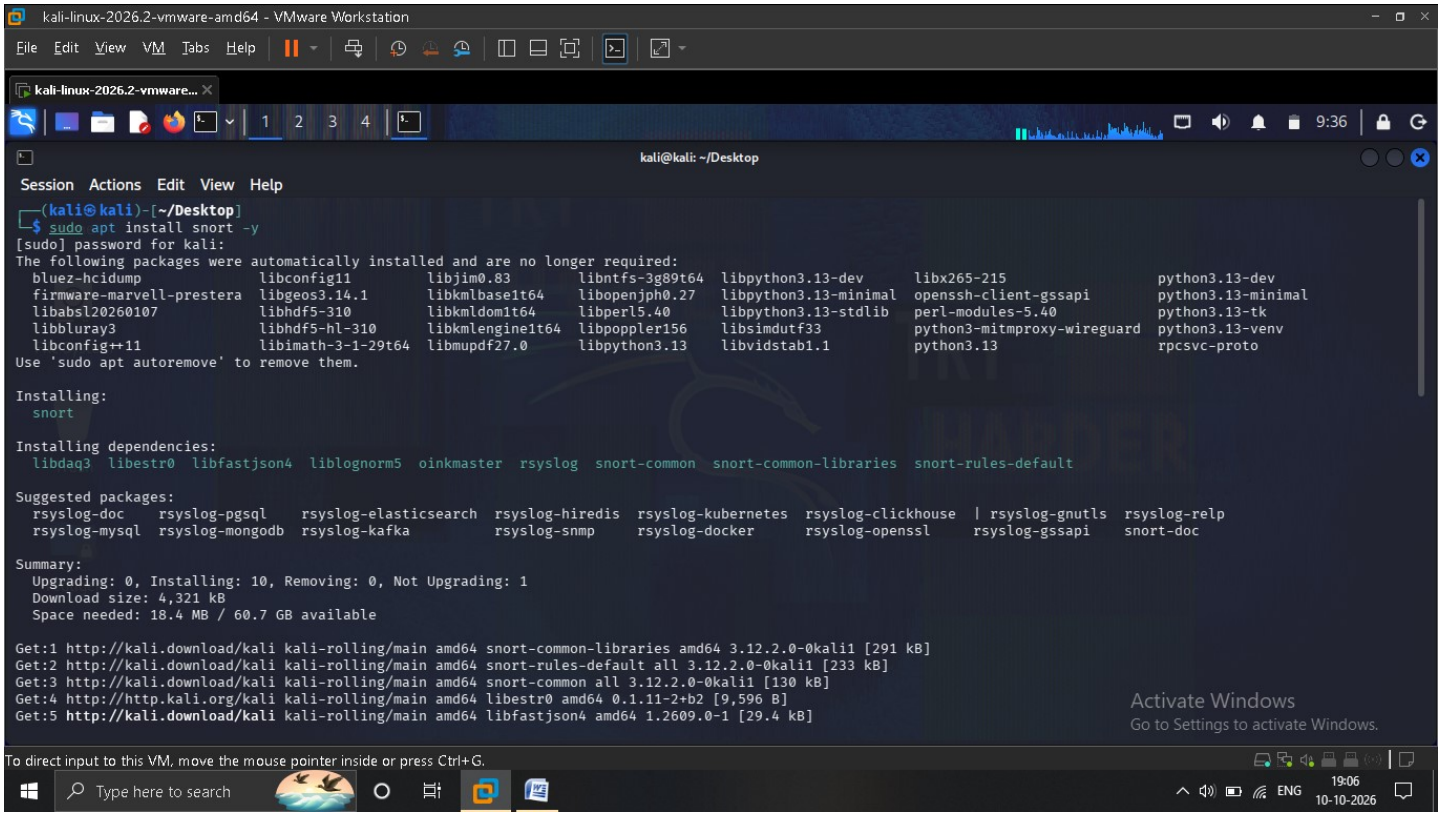
Terminal

An isolated or authorized lab network

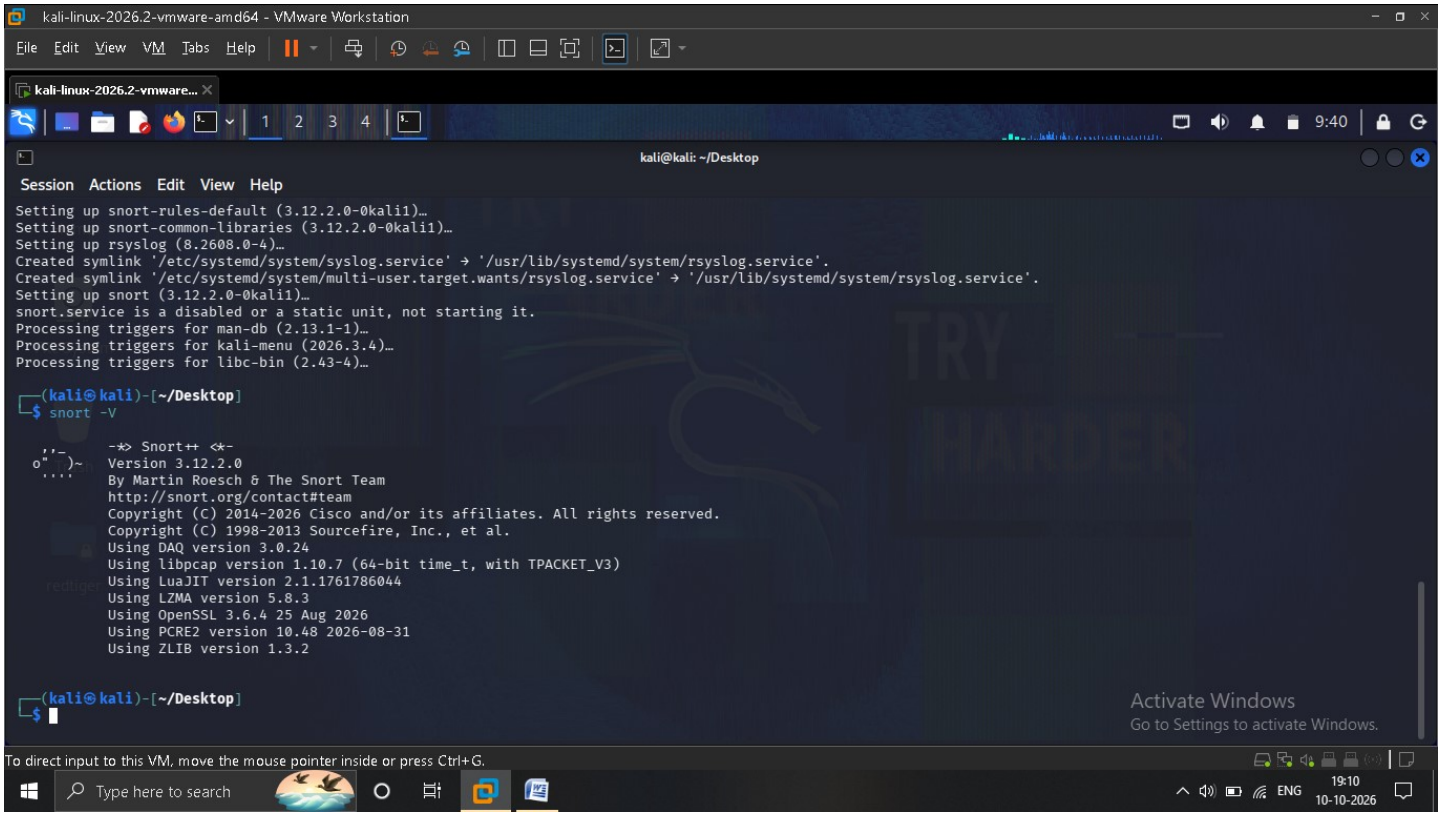
3.What Is Snort?

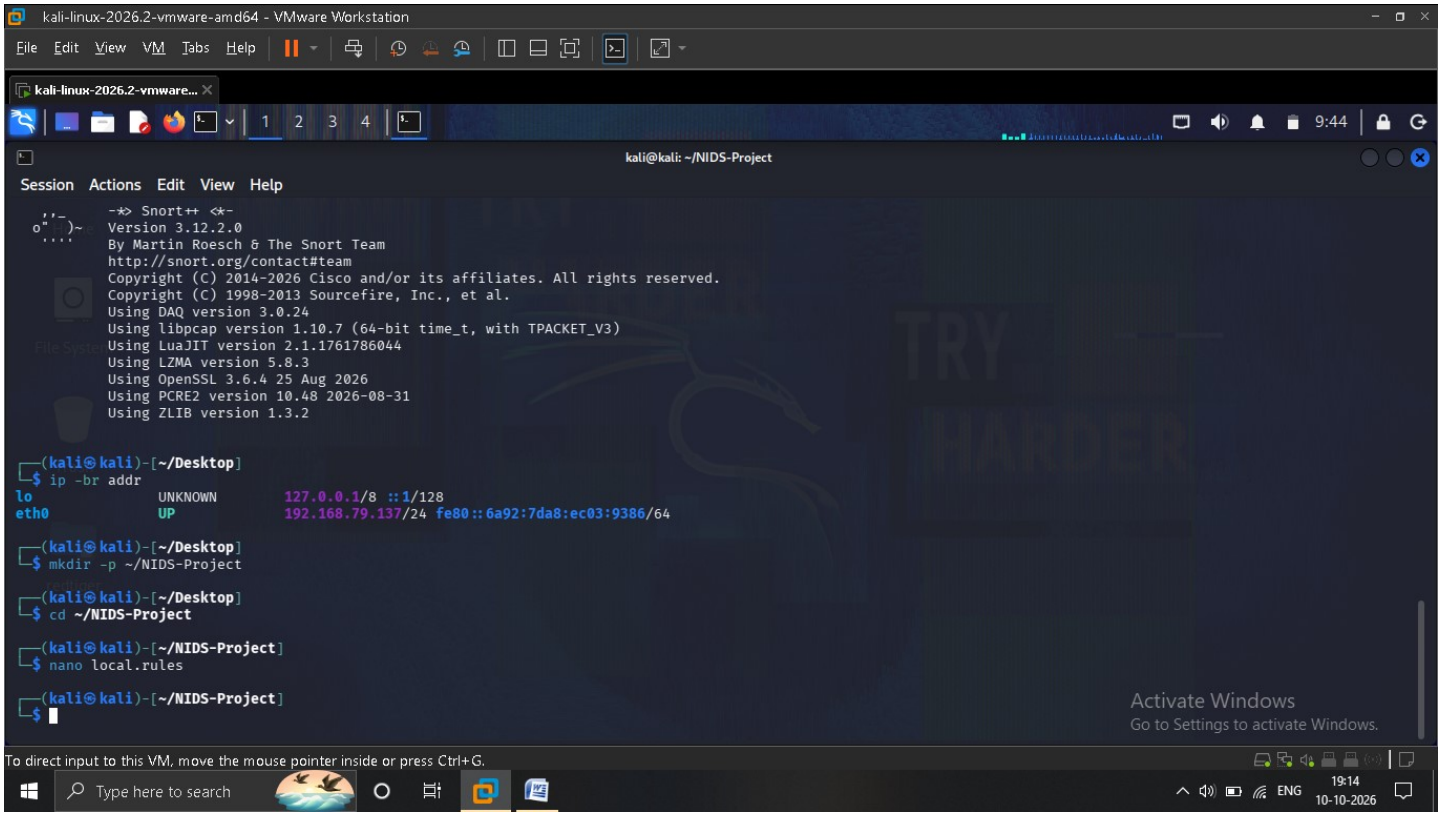
Snort is an open-source network security tool that can monitor and analyze network packets. It uses rules to identify selected traffic patternand can generate alerts when matching packets are detected.

Installing Snort Explanation: Snort was installed on Kali Linux using the terminal. Snort is used to monitor network traffic and detect suspicious activity.

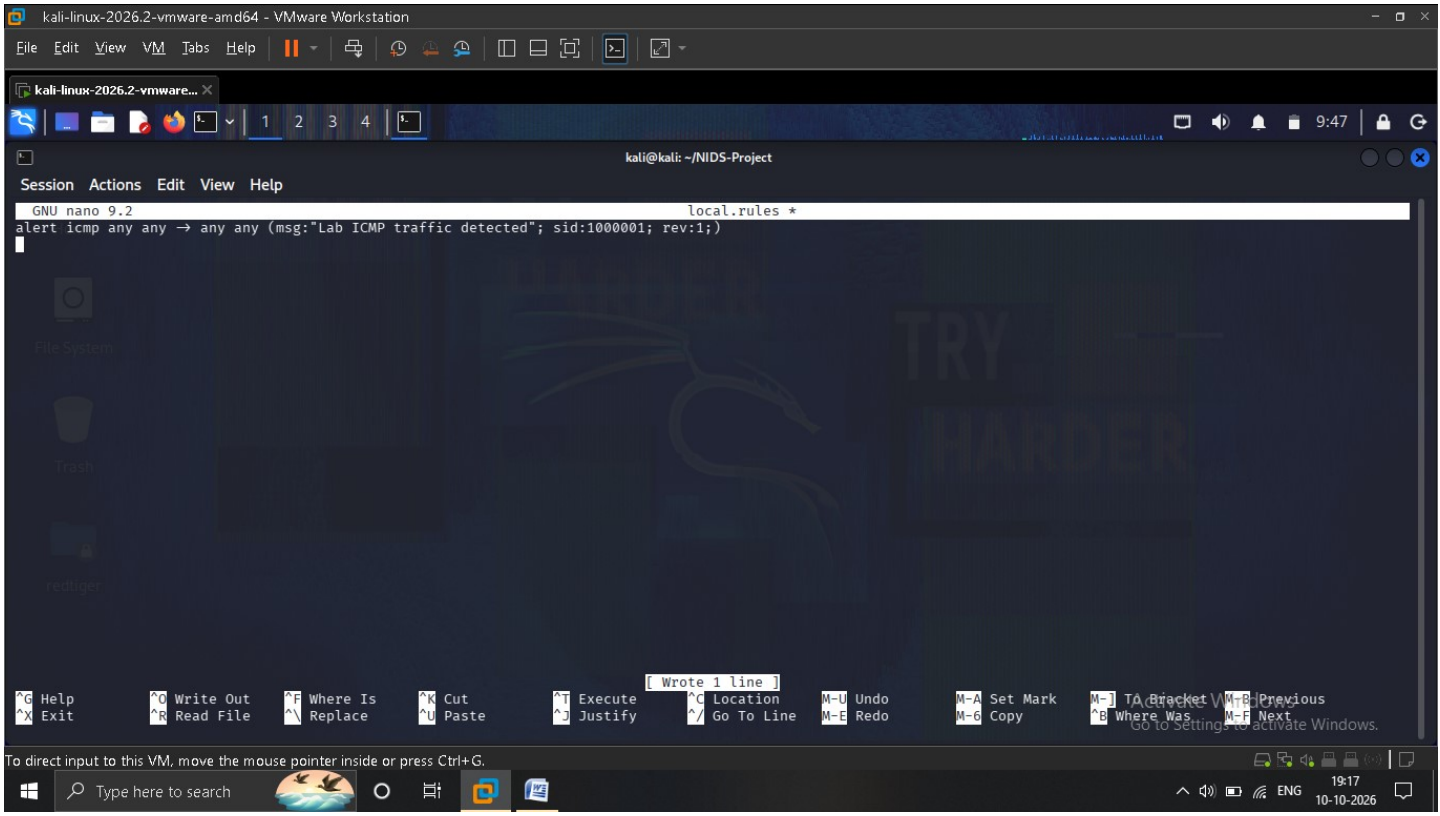


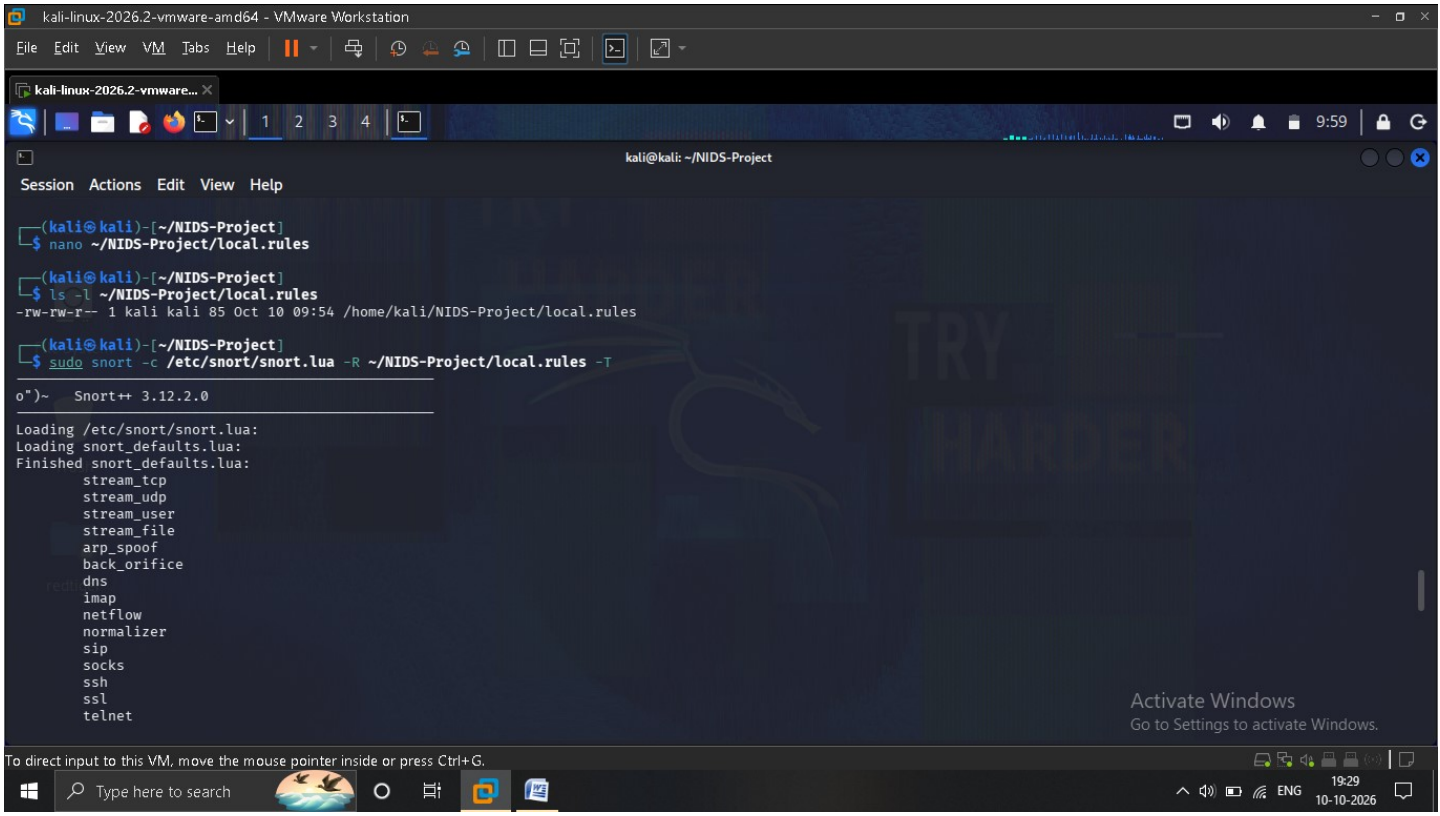
Checking the Snort Version Explanation: The Snort version was checked to confirm that the installation was successful. The installed version is Snort 3.12.2.0.

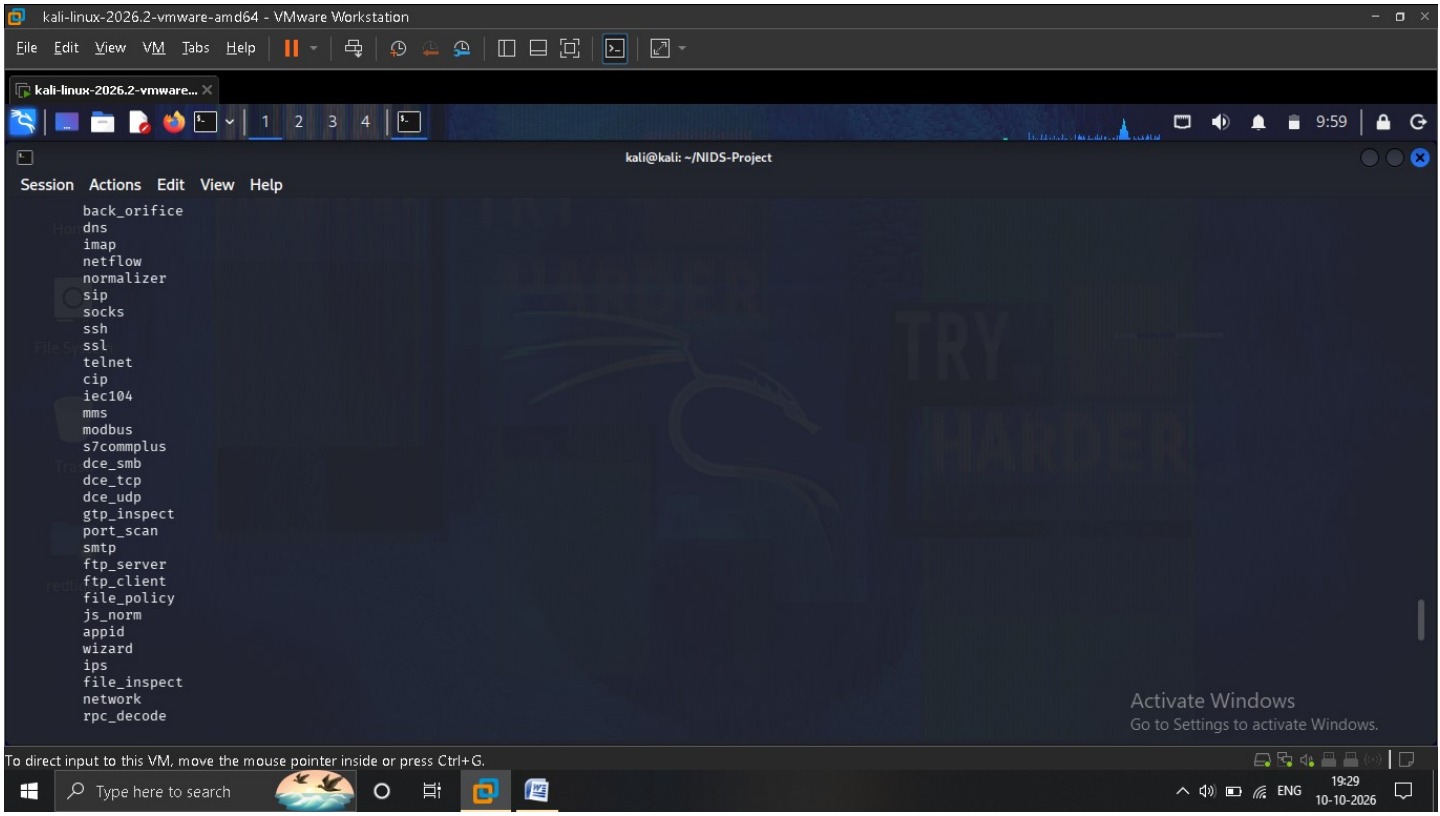

kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
9:40
kali@kali: ~/Desktop
Session Actions Edit View Help
Setting up snort-rules-default (3.12.2.0-0kali1)…
Setting up snort-common-libraries (3.12.2.0-0kali1)…
Setting up rsyslog (8.2608.0-4)…
Created symlink '/etc/systemd/system/syslog.service' → '/usr/lib/systemd/system/rsyslog.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/rsyslog.service' → '/usr/lib/systemd/system/rsyslog.service'.
Setting up snort (3.12.2.0-0kali1)…
snort.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1)…
Processing triggers for kali-menu (2026.3.4)…
Processing triggers for libc-bin (2.43-4)…
(kali㉿kali)-[~/Desktop]
$ snort -V
-*> Snort++ <*-
Version 3.12.2.0
By Martin Roesch & The Snort Team
http://snort.org/contact#team
Copyright (C) 2014-2026 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using DAQ version 3.0.24
Using libpcap version 1.10.7 (64-bit time_t, with TPACKET_V3)
Using LuaJIT version 2.1.1761786044
Using LZMA version 5.8.3
Using OpenSSL 3.6.4 25 Aug 2026
Using PCRE2 version 10.48 2026-08-31
Using ZLIB version 1.3.2
(kali㉿kali)-[~/Desktop]
$
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:10
10-10-2026

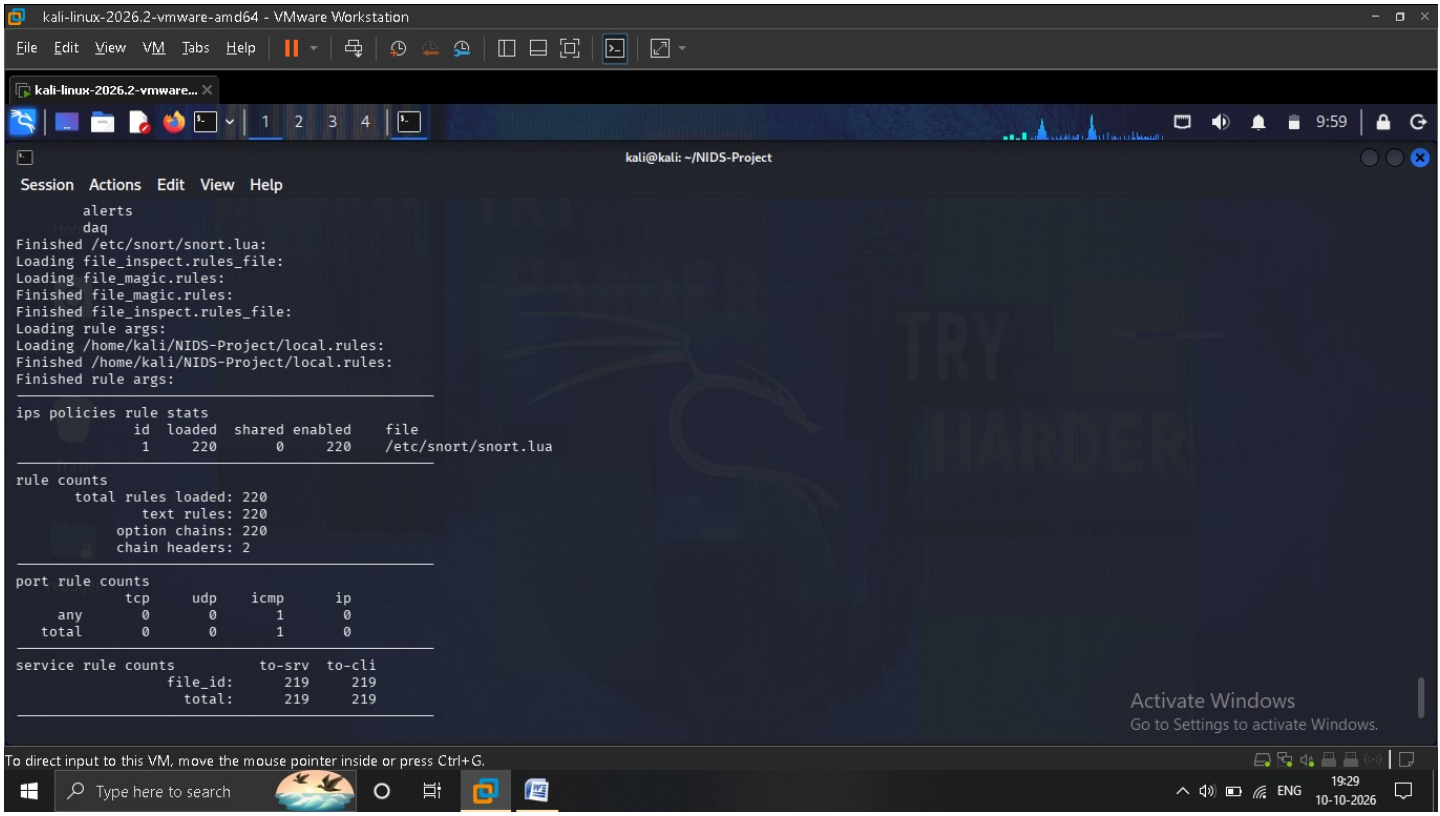


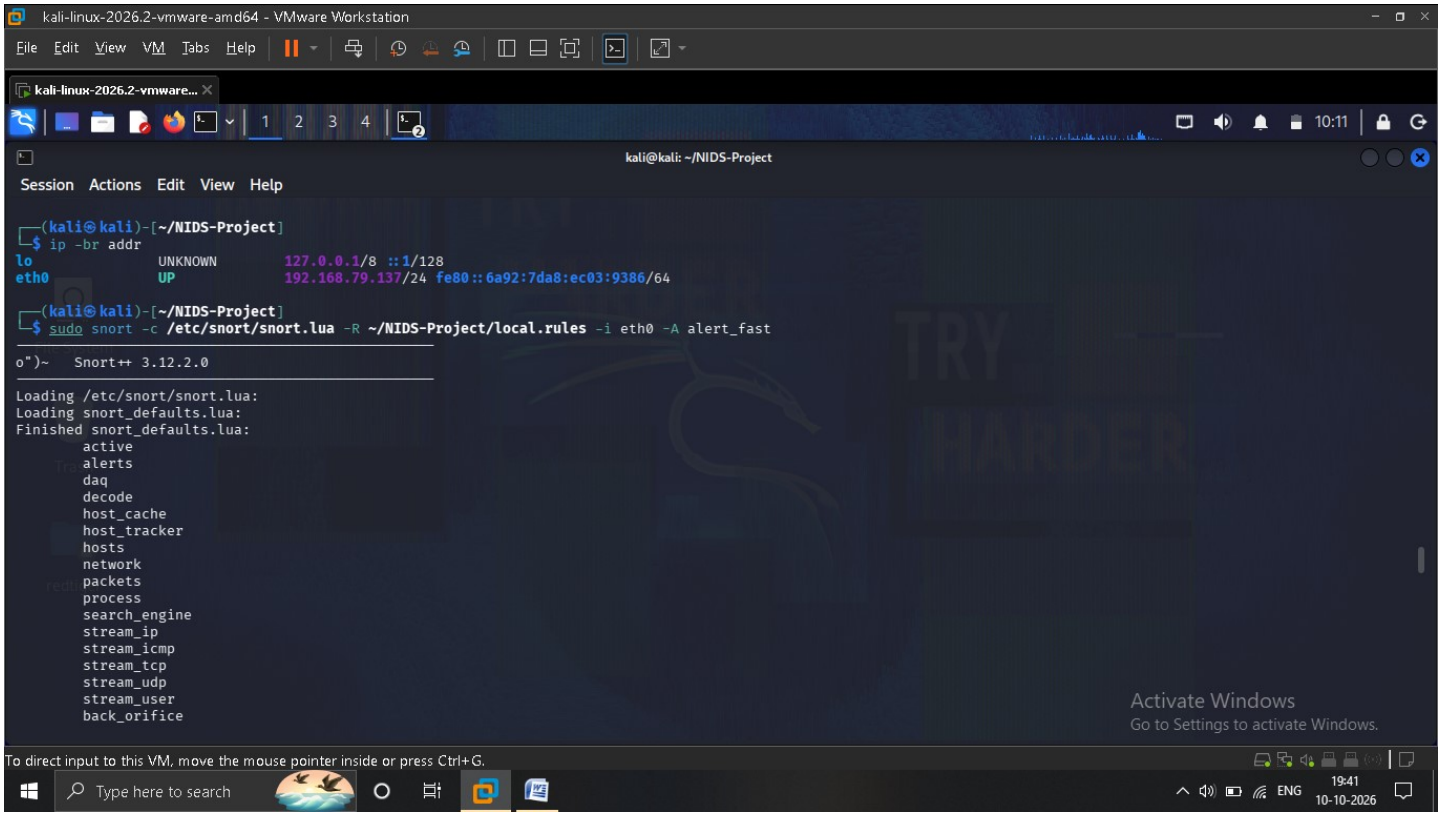
3. Creating the Detection Rule Explanation: A custom ICMP detection rule was created in the local.rules file. This rule generates an alert when matching ICMP traffic is detected.

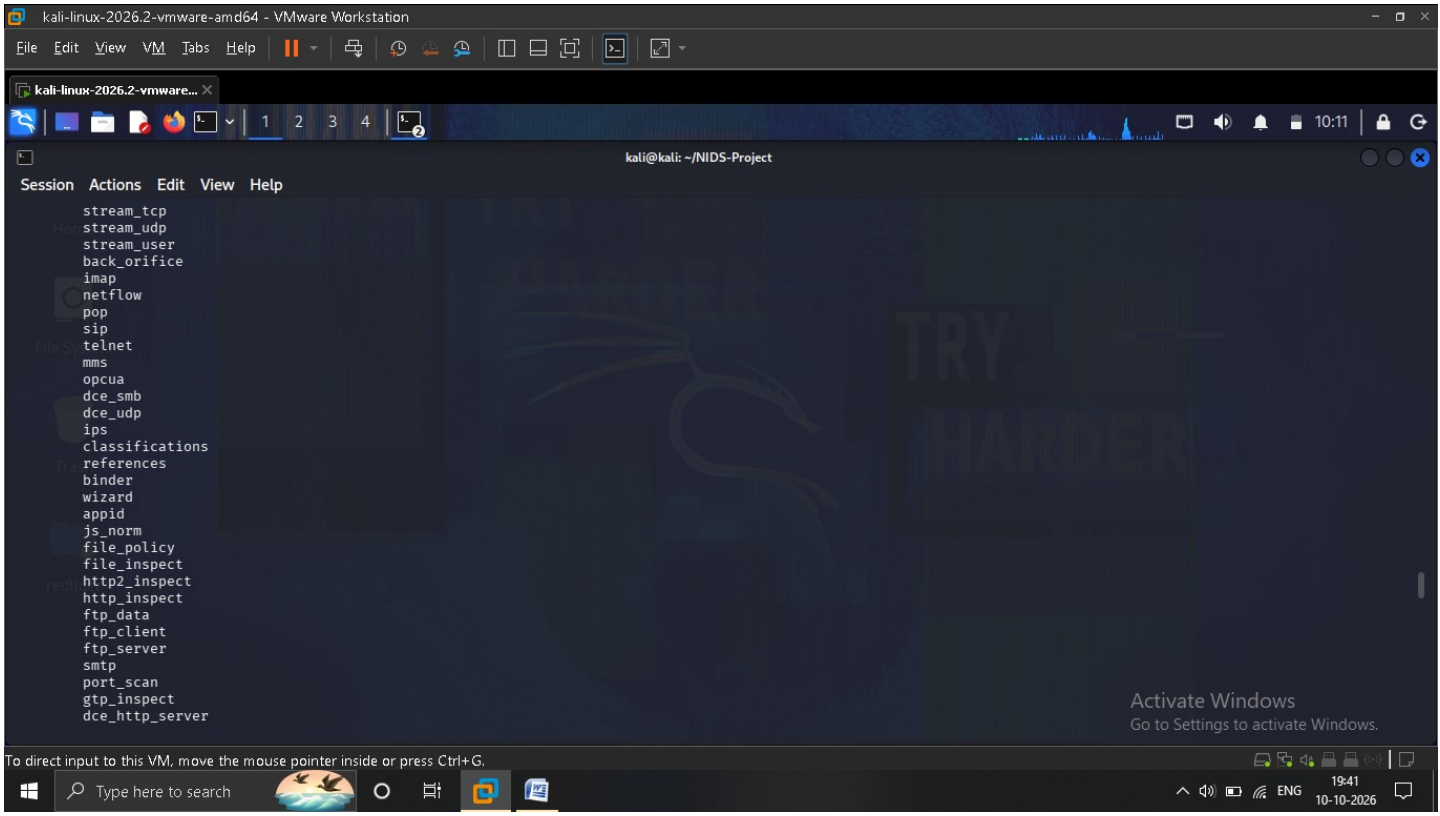

kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
9:47
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
GNU nano 9.2 local.rules *
alert icmp any any → any any (msg:"Lab ICMP traffic detected"; sid:1000001; rev:1;)
[Wrote 1 line]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-P Previous
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^B Where Was M-F Next
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:17
10-10-2026

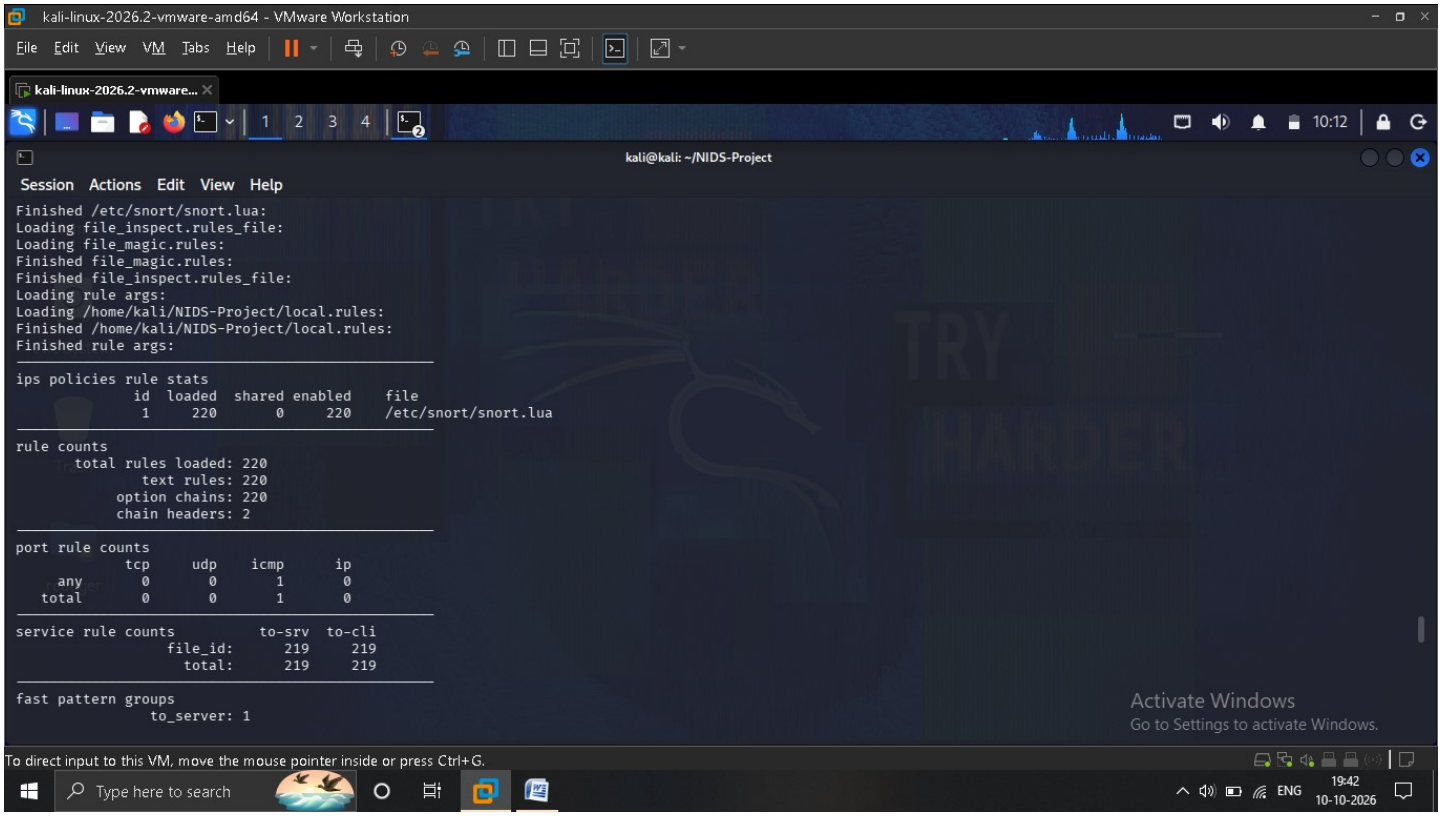

kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
(kali㉿kali)-[~/NIDS-Project]
$ nano ~/NIDS-Project/local.rules
(kali㉿kali)-[~/NIDS-Project]
$ ls -l ~/NIDS-Project/local.rules
-rw-rw-r-- 1 kali kali 85 Oct 10 09:54 /home/kali/NIDS-Project/local.rules
(kali㉿kali)-[~/NIDS-Project]
$ sudo snort -c /etc/snort/snort.lua -R ~/NIDS-Project/local.rules -T
o")~ Snort++ 3.12.2.0
Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
stream_tcp
stream_udp
stream_user
stream_file
arp_spoof
back_orifice
dns
imap
netflow
normalizer
sip
socks
ssh
ssl
telnet
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:29
10-10-2026


kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
9:59
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
back_orifice
dns
imap
netflow
normalizer
sip
socks
ssh
ssl
telnet
cip
iec104
mms
modbus
s7commplus
dce_smb
dce_tcp
dce_udp
gtp_inspect
port_scan
smtp
ftp_server
ftp_client
file_policy
js_norm
appid
wizard
ips
file_inspect
network
rpc_decode
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:29
10-10-2026


kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
9:59
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
alerts
daq
Finished /etc/snort/snort.lua:
Loading file_inspect.rules_file:
Loading file_magic.rules:
Finished file_magic.rules:
Finished file_inspect.rules_file:
Loading rule args:
Loading /home/kali/NIDS-Project/local.rules:
Finished /home/kali/NIDS-Project/local.rules:
Finished rule args:
ips policies rule stats
id loaded shared enabled file
1 220 0 220 /etc/snort/snort.lua
rule counts
total rules loaded: 220
text rules: 220
option chains: 220
chain headers: 2
port rule counts
tcp udp icmp ip
any 0 0 1 0
total 0 0 1 0
service rule counts to-srv to-cli
file_id: 219 219
total: 219 219
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:29
10-10-2026

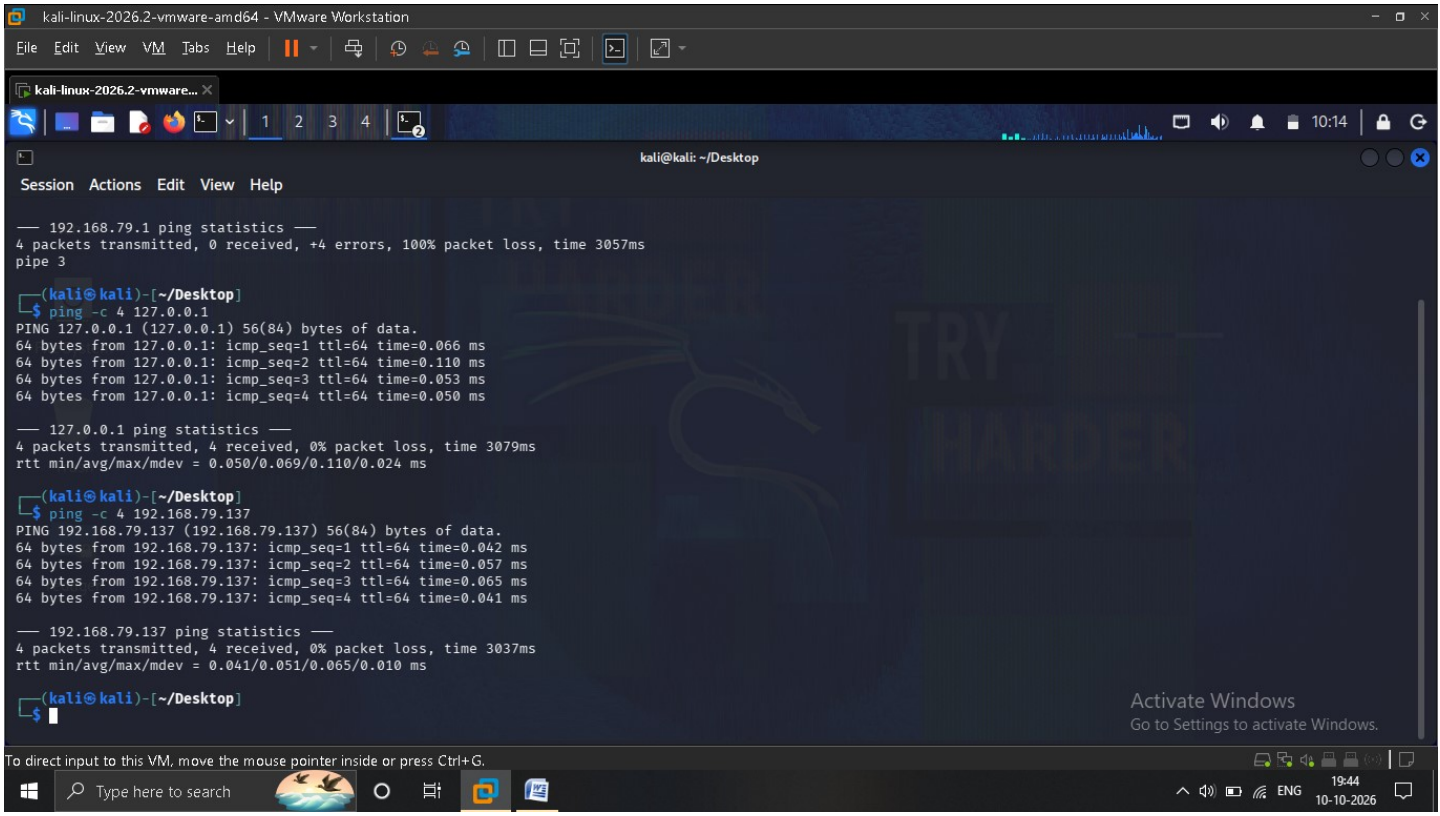

kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
10:11
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
(kali㉿kali)-[~/NIDS-Project]
$ ip -br addr
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.79.137/24 fe80::6a92:7da8:ec03:9386/64
(kali㉿kali)-[~/NIDS-Project]
$ sudo snort -c /etc/snort/snort.lua -R ~/NIDS-Project/local.rules -i eth0 -A alert_fast
o")~ Snort++ 3.12.2.0
Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
active
alerts
daq
decode
host_cache
host_tracker
hosts
network
packets
process
search_engine
stream_ip
stream_icmp
stream_tcp
stream_udp
stream_user
back_orifice
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:41
10-10-2026


kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
10:11
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
stream_tcp
stream_udp
stream_user
back_orifice
imap
netflow
pop
sip
telnet
mms
opcua
dce_smb
dce_udp
ips
classifications
references
binder
wizard
appid
js_norm
file_policy
file_inspect
http2_inspect
http_inspect
ftp_data
ftp_client
ftp_server
smtp
port_scan
gtp_inspect
dce_http_server
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:41
10-10-2026

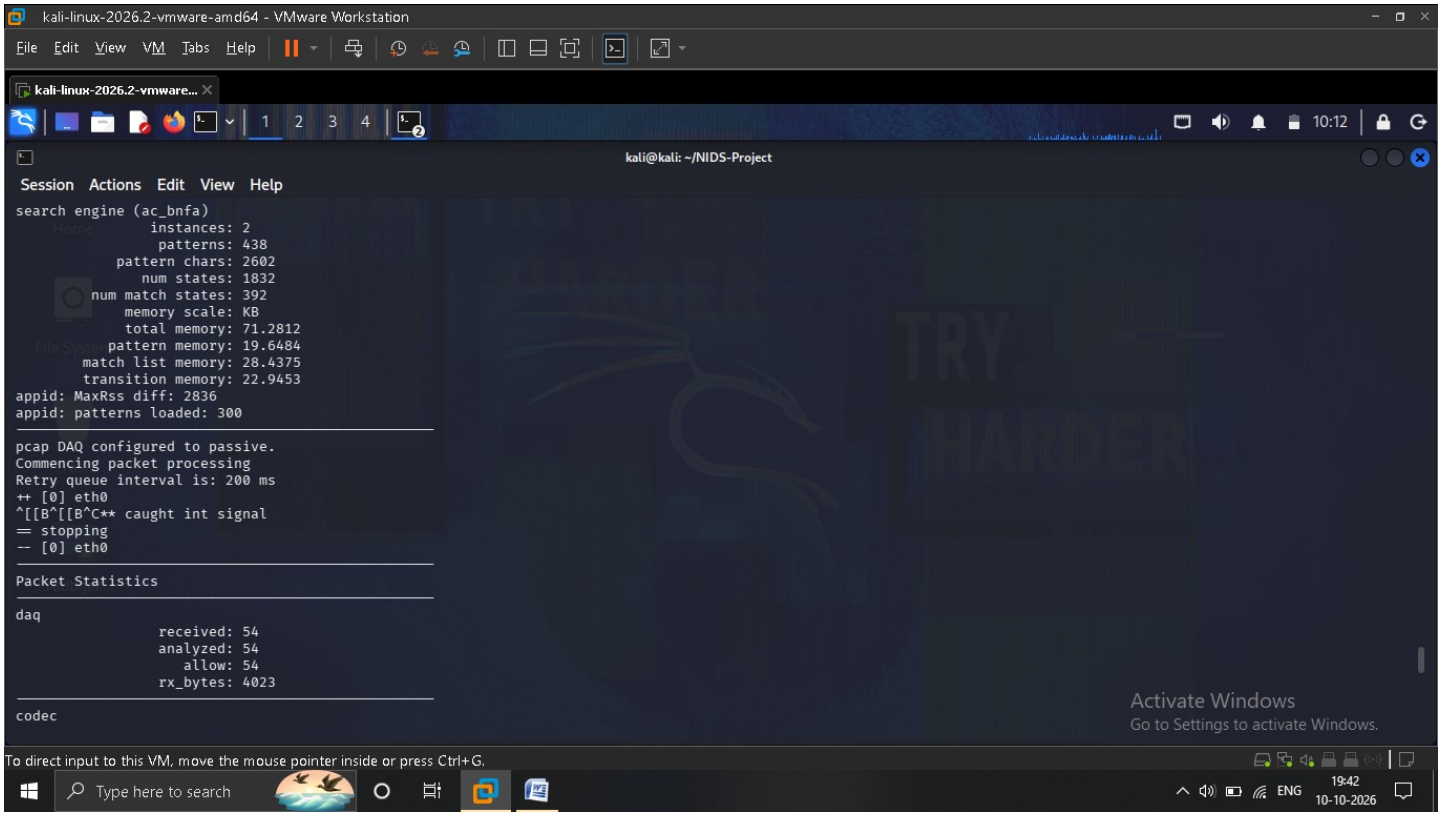


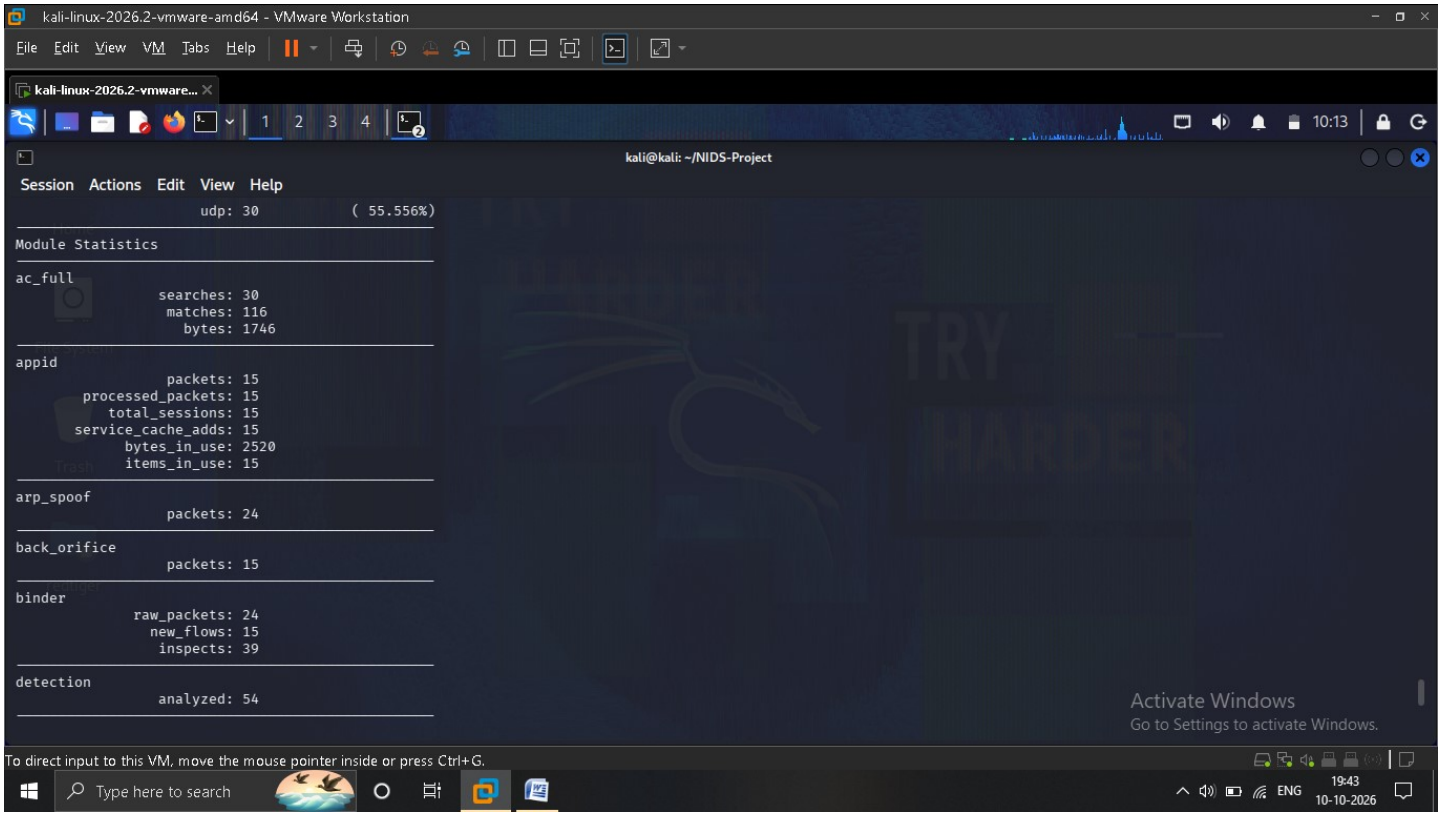
Second Terminal

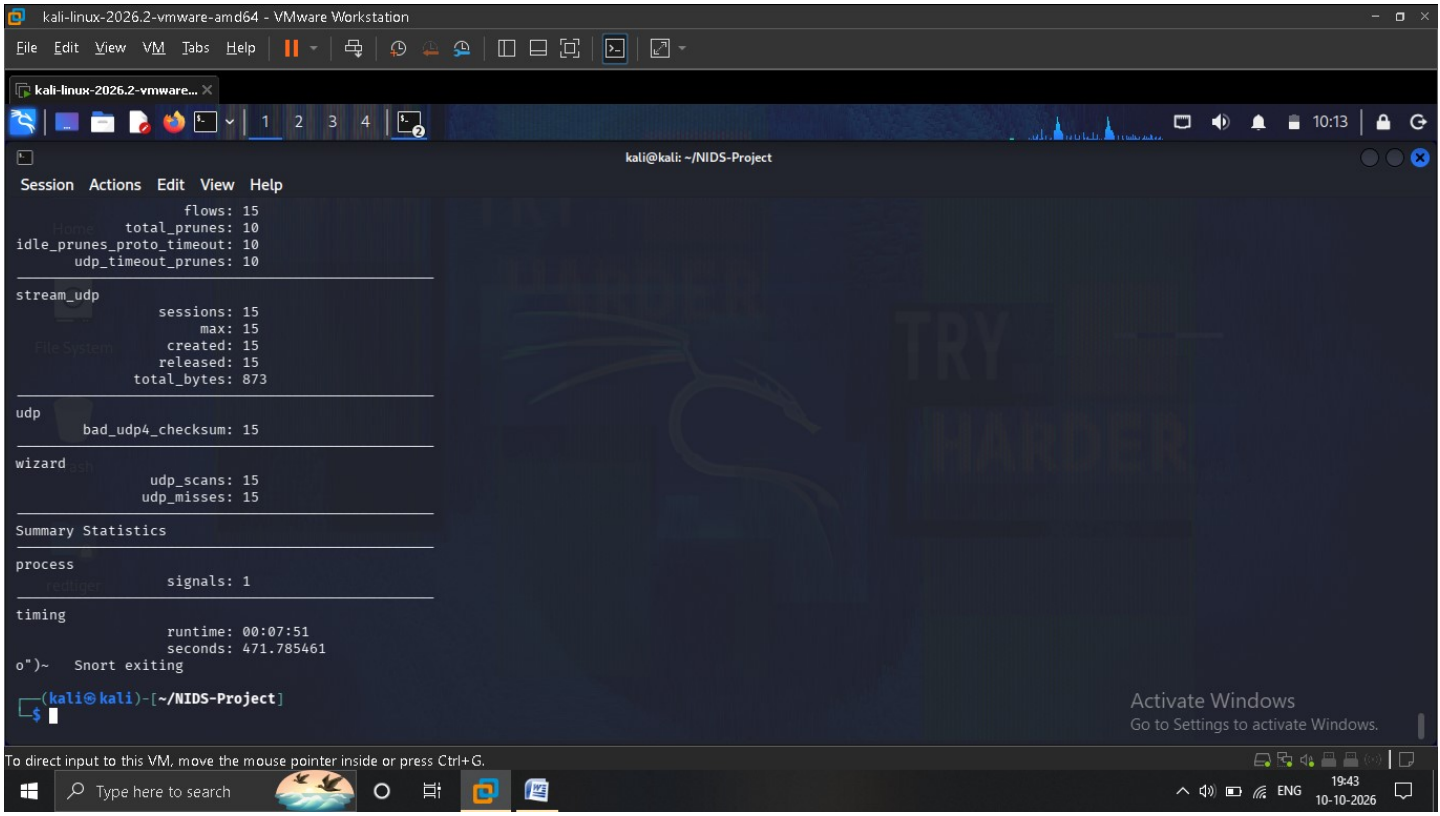
Explanation: Test traffic was generated using the ping command. Snort detected matching ICMP packets and displayed alerts with the message “Lab ICMP traffic detected”.



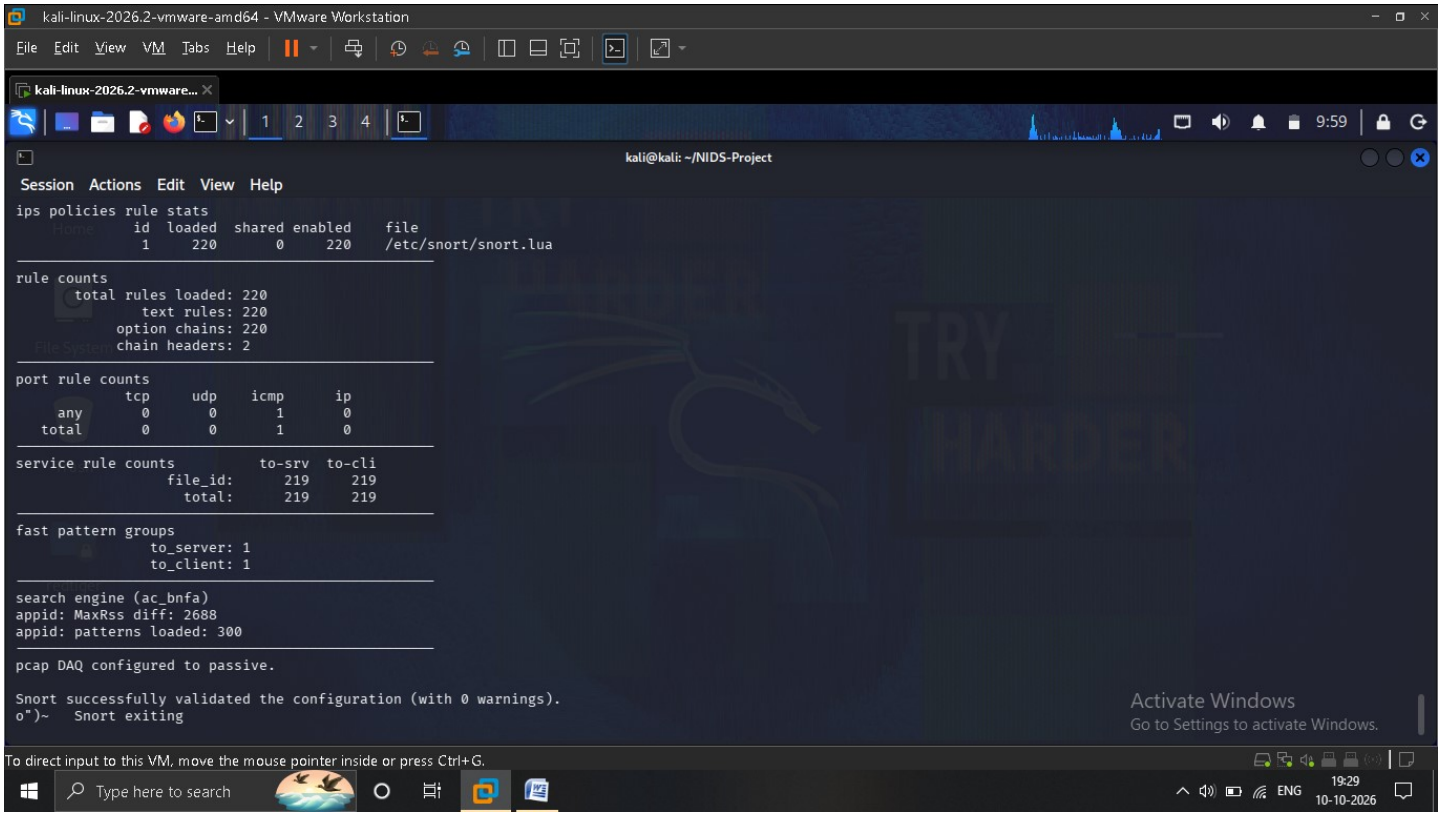
Back to first terminal

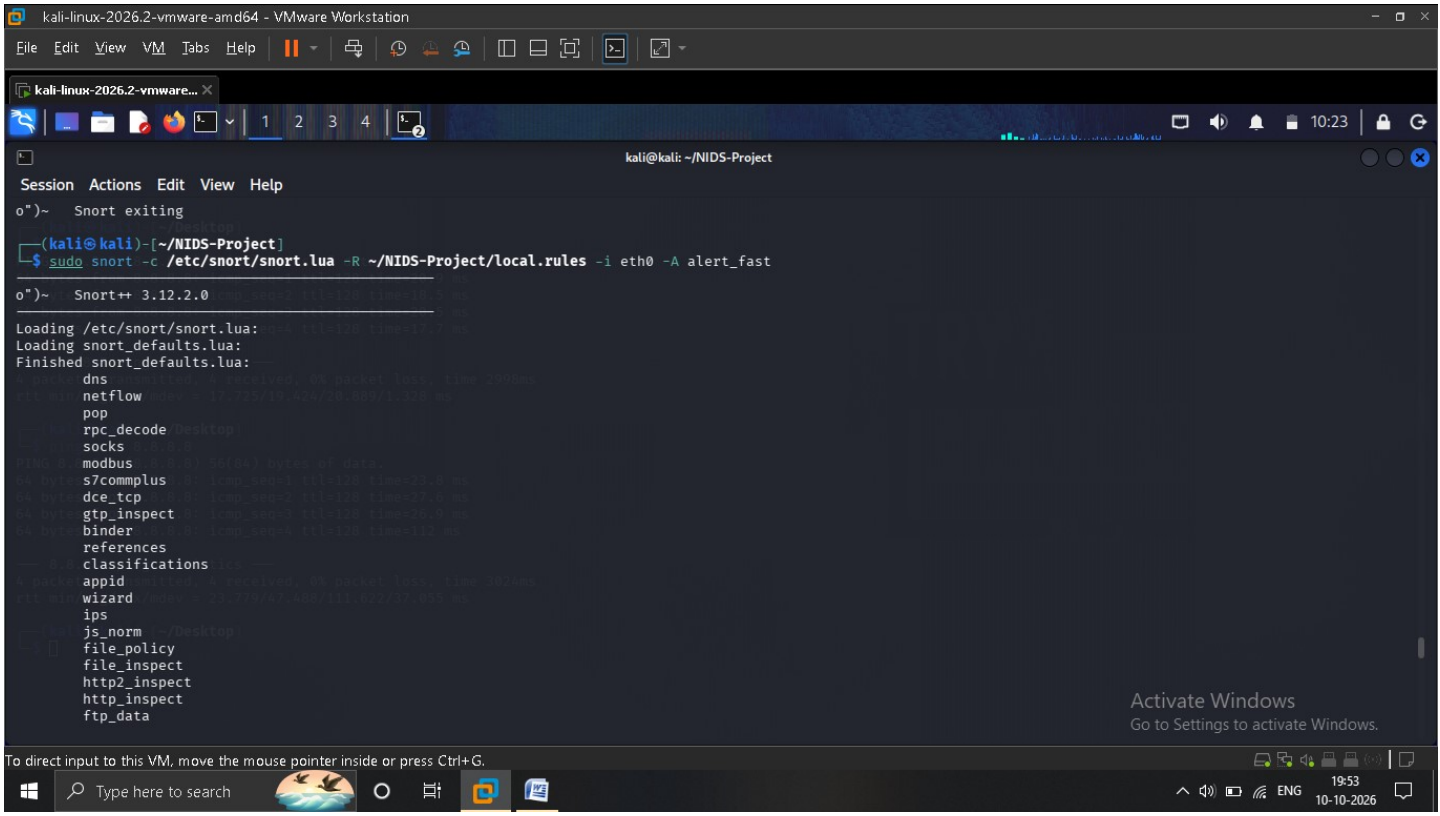

kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
10:12
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
search engine (ac_bnfa)
instances: 2
patterns: 438
pattern chars: 2602
num states: 1832
num match states: 392
memory scale: KB
total memory: 71.2812
pattern memory: 19.6484
match list memory: 28.4375
transition memory: 22.9453
appid: MaxRss diff: 2836
appid: patterns loaded: 300
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
^[[B^[[B^C** caught int signal
== stopping
-- [0] eth0
Packet Statistics
daq
received: 54
analyzed: 54
allow: 54
rx_bytes: 4023
codec
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:42
10-10-2026

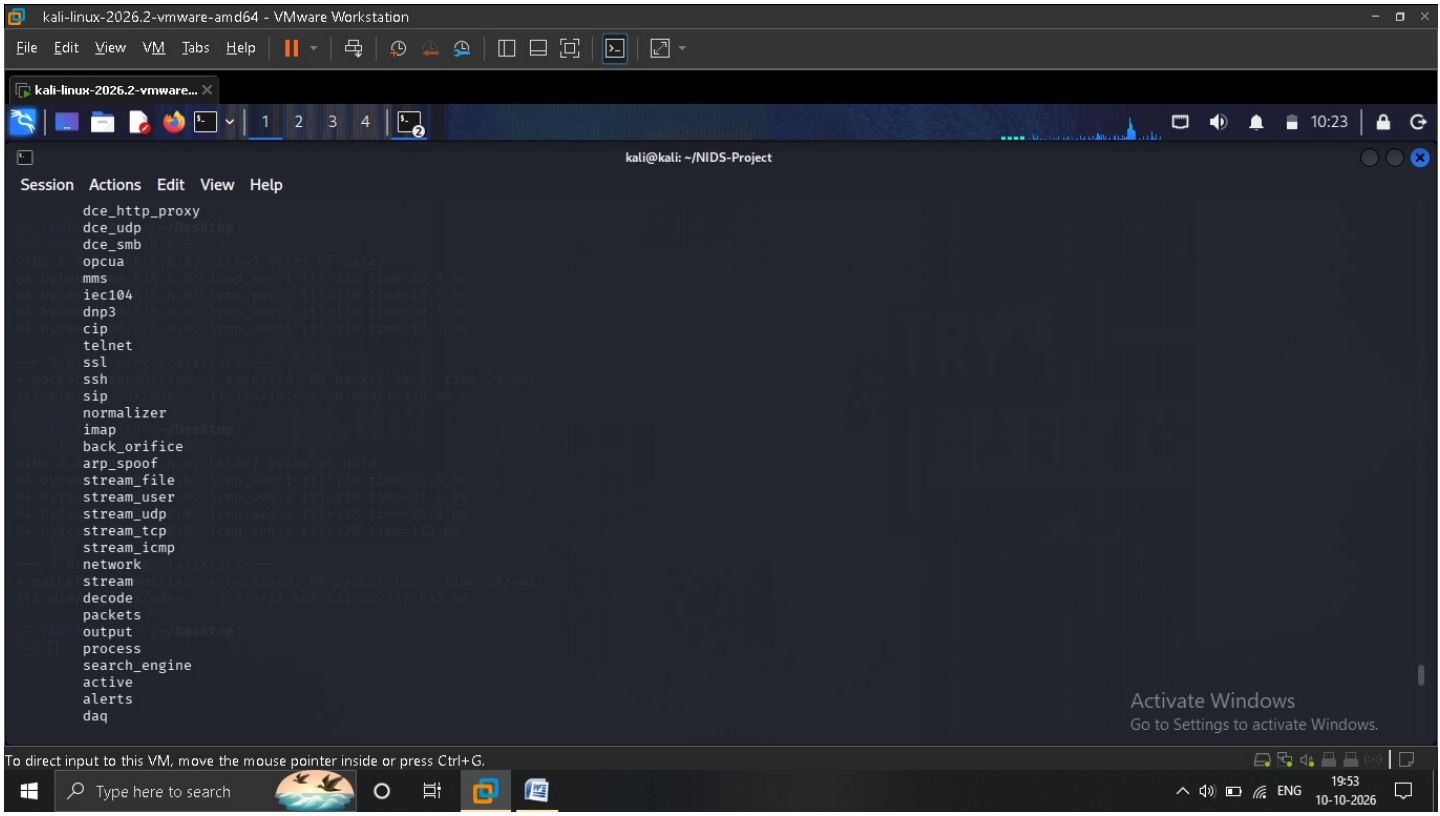

kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
udp: 30 (55.556%)
Module Statistics
ac_full
searches: 30
matches: 116
bytes: 1746
appid
packets: 15
processed_packets: 15
total_sessions: 15
service_cache_adds: 15
bytes_in_use: 2520
items_in_use: 15
arp_spoof
packets: 24
back_orifice
packets: 15
binder
raw_packets: 24
new_flows: 15
inspects: 39
detection
analyzed: 54
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:43
10-10-2026

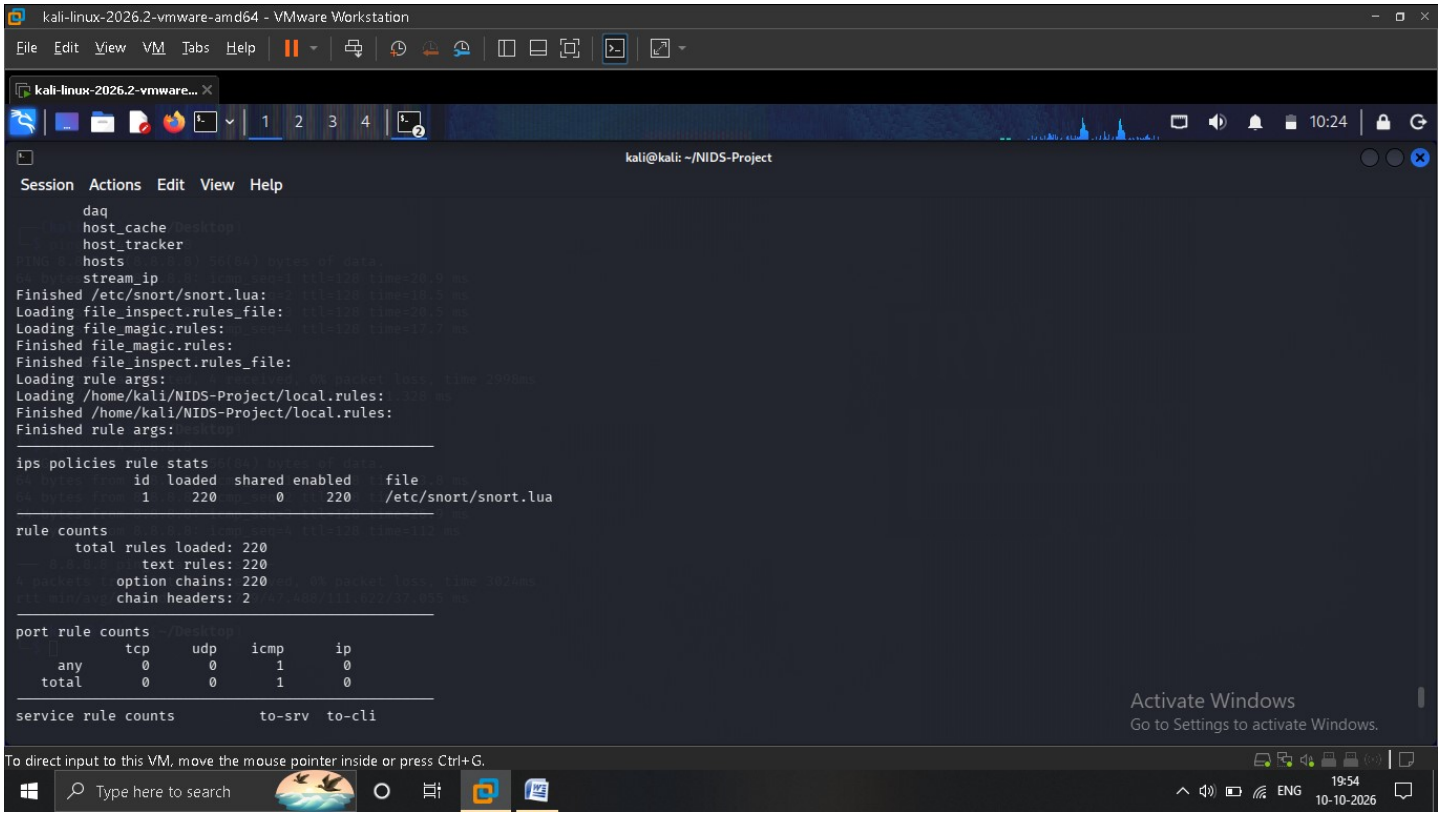


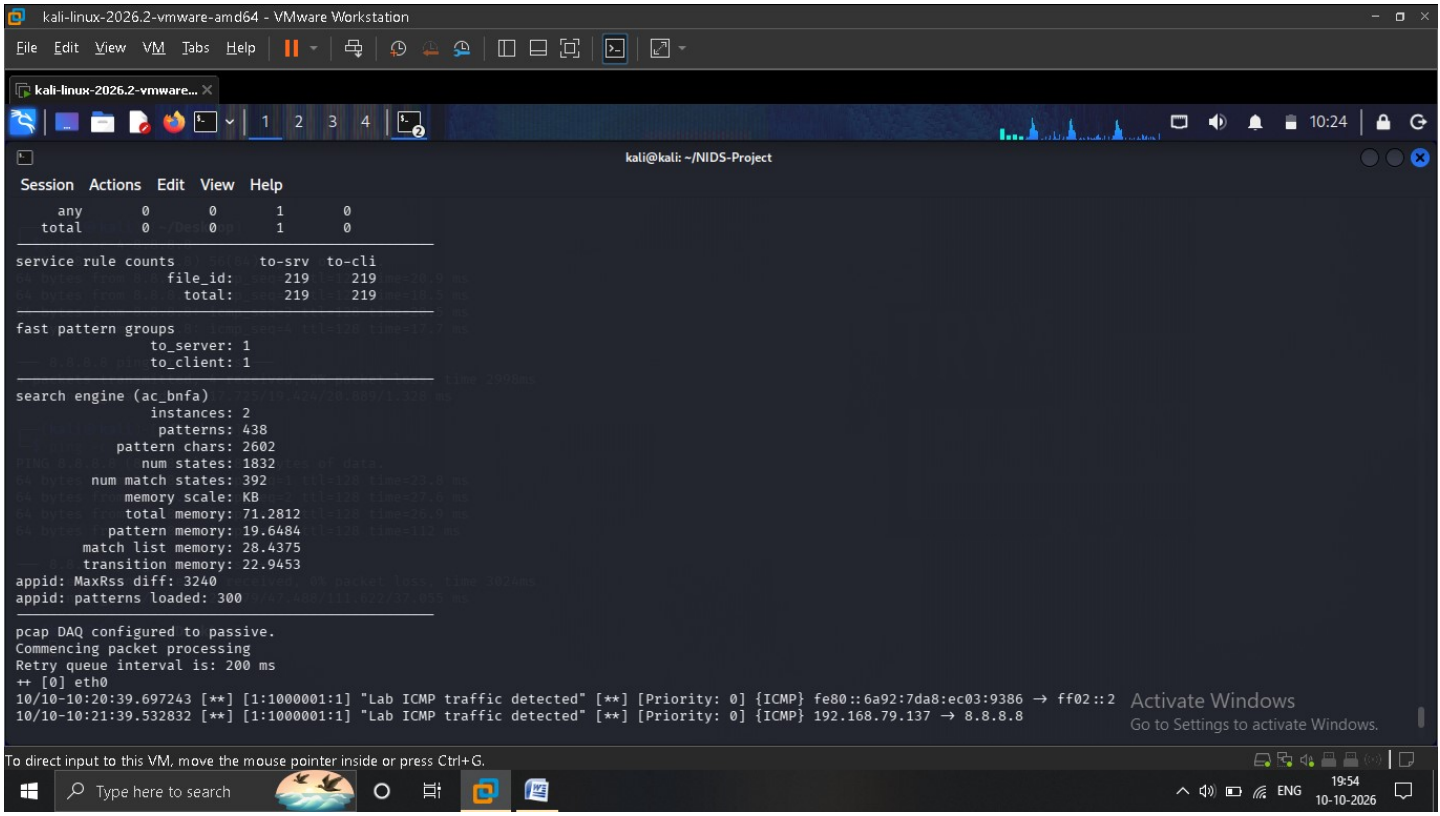
Testing Results: Snort 3.12.2.0 was configured on Kali Linux and started monitoring the `eth0` network interface. The custom ICMP rule loaded successfully. During the monitoring session, Snort received and analyzed 54 packets. No matching custom-rule alert was confirmed in the observed output.


kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
ips policies rule stats
id loaded shared enabled file
1 220 0 220 /etc/snort/snort.lua
rule counts
total rules loaded: 220
text rules: 220
option chains: 220
chain headers: 2
port rule counts
tcp udp icmp ip
any 0 0 1 0
total 0 0 1 0
service rule counts to-srv to-cli
file_id: 219 219
total: 219 219
fast pattern groups
to_server: 1
to_client: 1
search engine (ac_bnfa)
appid: MaxRss diff: 2688
appid: patterns loaded: 300
pcap DAQ configured to passive.
Snort successfully validated the configuration (with 0 warnings).
o")~ Snort exiting
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:29
10-10-2026

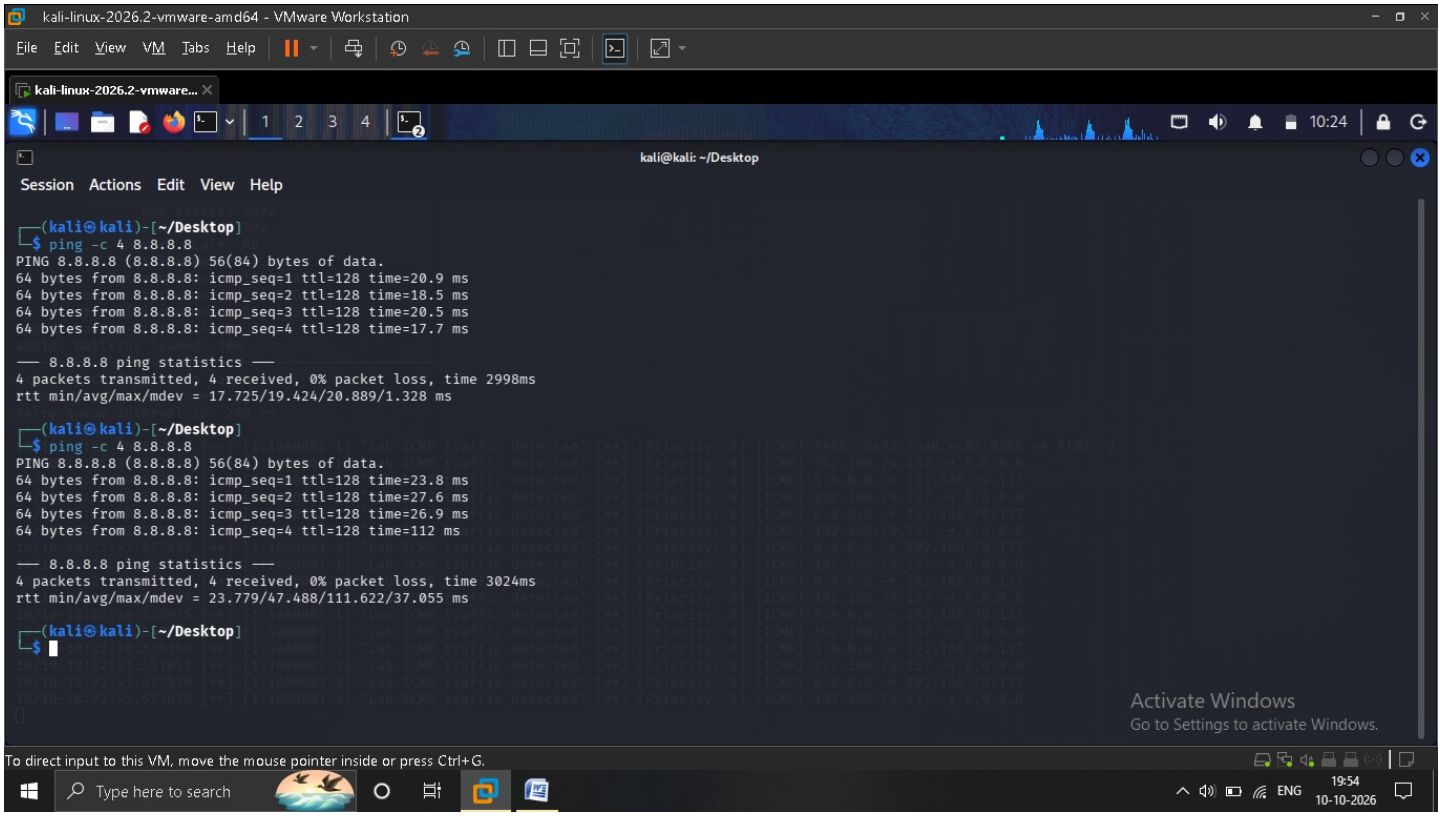

kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
10:23
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
o")~ Snort exiting
(kali㉿kali)-[~/NIDS-Project]
$ sudo snort -c /etc/snort/snort.lua -R ~/NIDS-Project/local.rules -i eth0 -A alert_fast
o")~ Snort++ 3.12.2.0
Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
dns
netflow
pop
rpc_decode
socks
modbus
s7commplus
dce_tcp
gtp_inspect
binder
references
classifications
appid
wizard
ips
js_norm
file_policy
file_inspect
http2_inspect
http_inspect
ftp_data
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:53
10-10-2026


kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
10:23
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
dce_http_proxy
dce_udp
dce_smb
opcua
mms
iec104
dnp3
cip
telnet
ssl
ssh
sip
normalizer
imap
back_orifice
arp_spoof
stream_file
stream_user
stream_udp
stream_tcp
stream_icmp
network
stream
decode
packets
output
process
search_engine
active
alerts
daq
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:53
10-10-2026


kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.2-vmware...
1 2 3 4
10:24
kali@kali: ~/NIDS-Project
Session Actions Edit View Help
daq
host_cache
host_tracker
hosts
stream_ip
Finished /etc/snort/snort.lua:
Loading file_inspect.rules_file:
Loading file_magic.rules:
Finished file_magic.rules:
Finished file_inspect.rules_file:
Loading rule args:
Loading /home/kali/NIDS-Project/local.rules:
Finished /home/kali/NIDS-Project/local.rules:
Finished rule args:
ips policies rule stats
id loaded shared enabled file
1 220 0 220 /etc/snort/snort.lua
rule counts
total rules loaded: 220
text rules: 220
option chains: 220
chain headers: 2
port rule counts
tcp udp icmp ip
any 0 0 1 0
total 0 0 1 0
service rule counts to-srv to-cli
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
19:54
10-10-2026



2nd Terminal



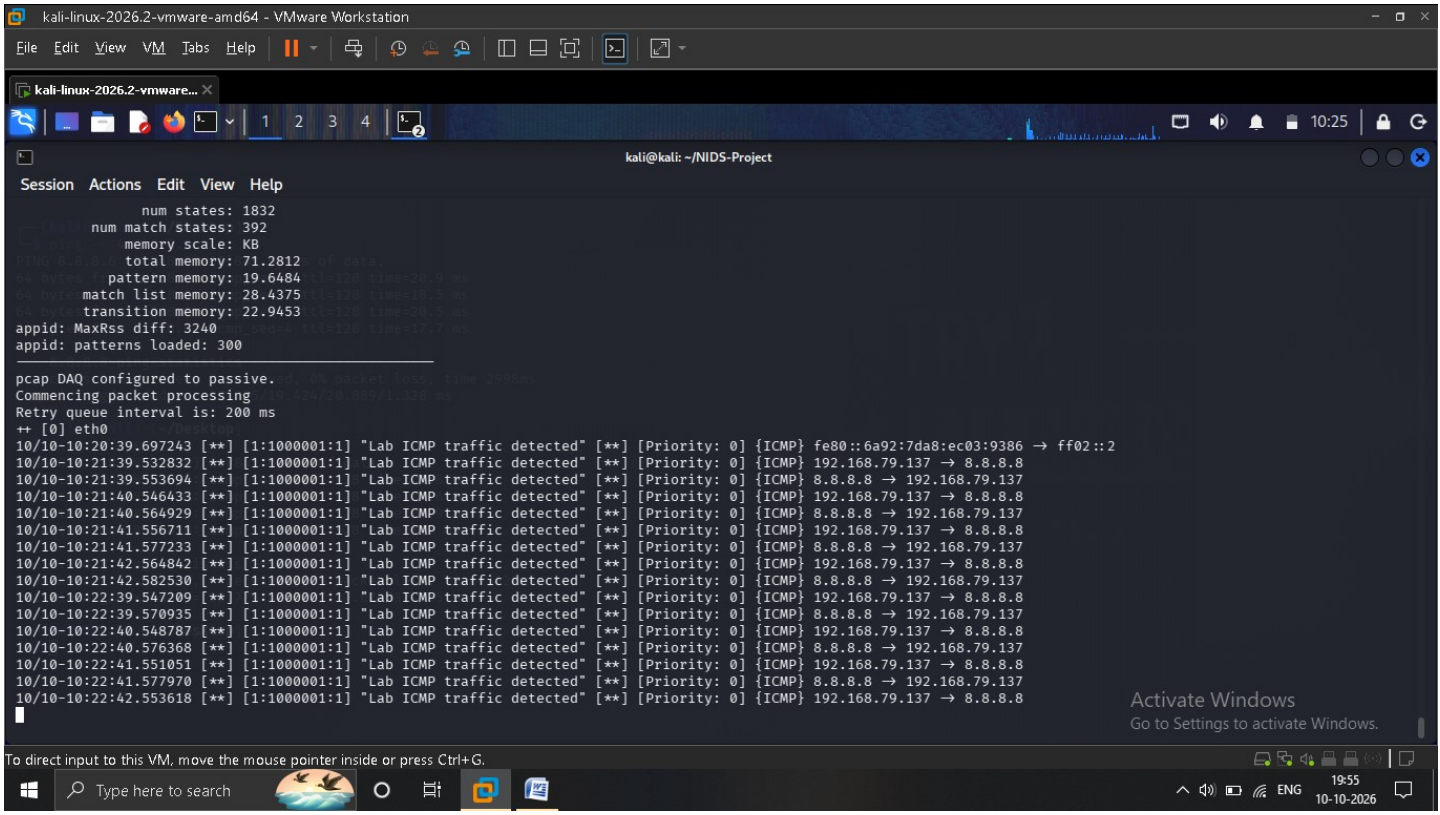
First Terminal

6. Testing and Results

Snort 3.12.2.0 was configured on Kali Linux and monitored network traffic through the `eth0` interface. The custom ICMP detection rule was loaded successfully.

During testing, Snort generated alerts with the message **"Lab ICMP traffic detected"** for ICMP packets observed between `192.168.79.137` and `8.8.8.8`.

Result: The custom ICMP detection rule successfully detected matching network traffic and generated alerts.



Conclusion

The project demonstrated the basic working of a Network Intrusion Detection System using Snort on Kali Linux. A custom ICMP rule was configured and successfully generated alerts for matching network traffic